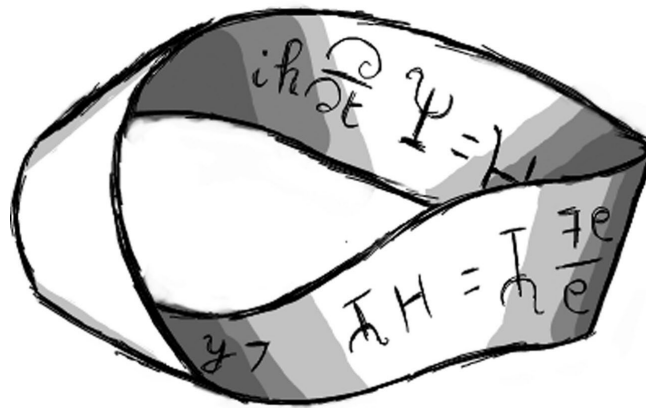
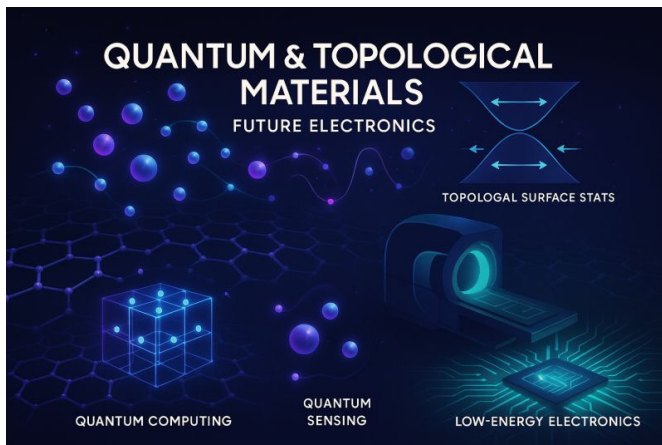


Deep Learning Electron Topology:

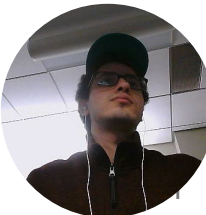
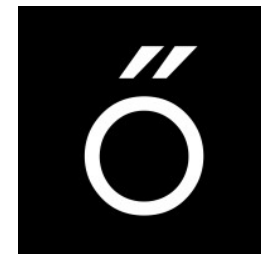
Predicting Materials with Topological Electronic Band Structures from only Stoichiometry.

Ahmed Fahmy

The Ohio State University

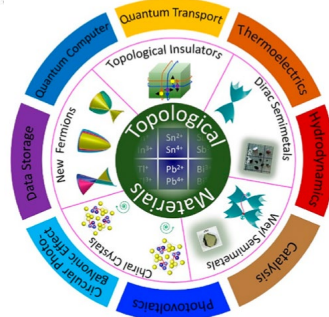
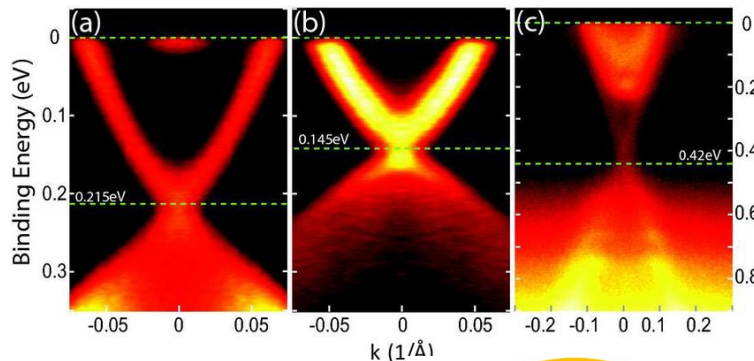
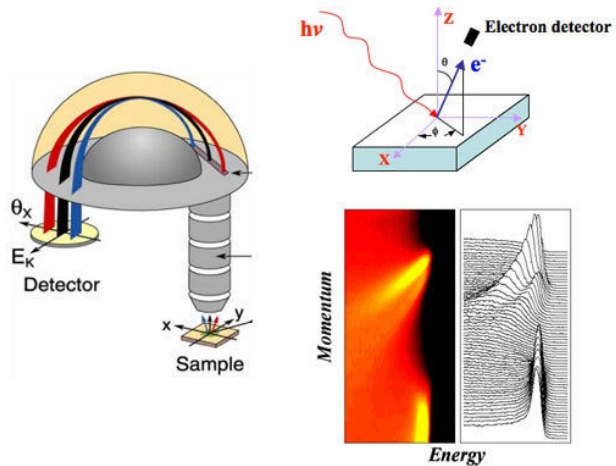


Fall 2025 Data Science Bootcamp
Erdos Institute



Predicting nontrivial topological electronic phases is essential for quantum technology.

Schematic Angle-Resolved Photoemission Spectroscopy (ARPES) ARPES probing topological surface states in bismuth selenide



ARPES & Scanning Tunneling Spectroscopy & Transport probes

✓ High accuracy

✗ Expensive, slow, and not scalable to large datasets

DFT Calculations

✗ Computationally costly

✗ cannot guaranteed to always give accurate results

- Their electronic behavior is determined not only by atomic composition and crystal symmetry, but by **global topological invariants** of the band structure.
- For example, topo insulators are **insulating in the bulk**, yet host **metallic conduction channels** on their surfaces or edges (bulk–boundary correspondence).
- These boundary states are **protected by symmetry** (e.g., time-reversal symmetry) and are **robust against disorder, impurities, and structural defects**.
- Topological materials offer promising applications in **spintronics, low-dissipation electronics, and topological quantum computing**.
- Topological phases arise from **spin–orbit coupling**, band inversion, and symmetry constraints — not easily inferred from conventional chemical intuition.

Recent literature

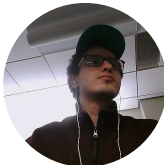
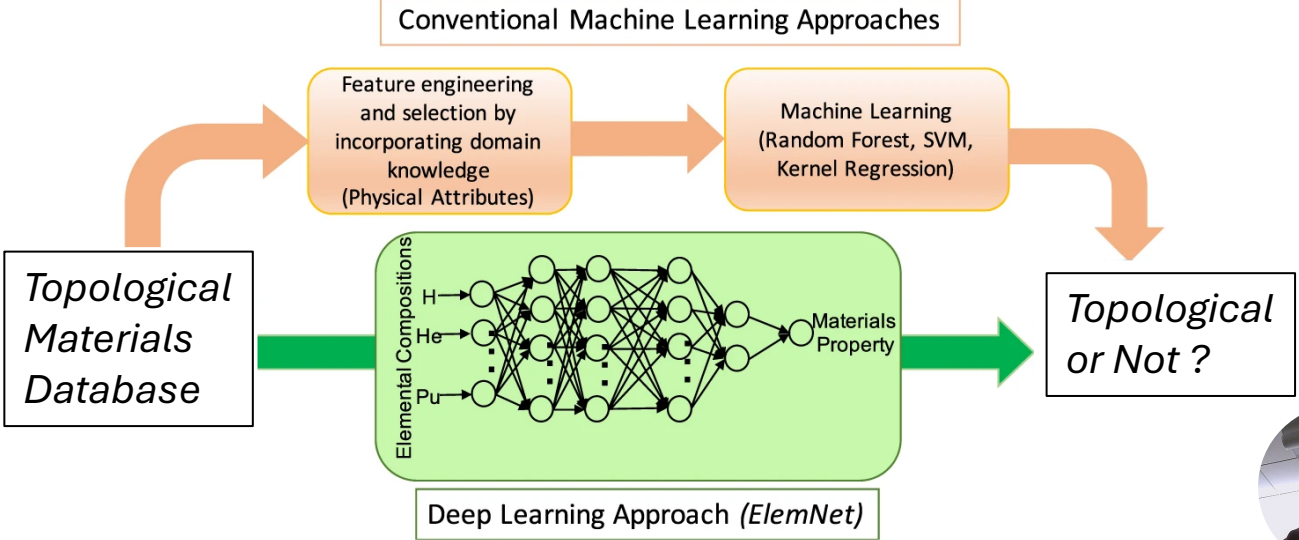
Input Descriptors:

- Chemical
- Structural
- Electronic

Train ML models on *Topological Materials ‘Materiae’* Database

Topological vs. Trivial

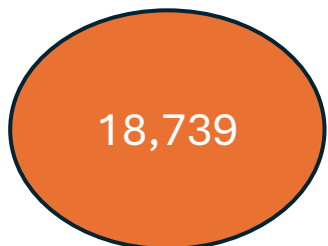
Confirm newly-found materials with *ab-initio* calculations



Data Collecting and preprocessing

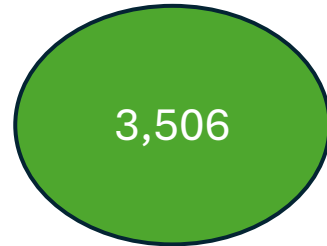
- **Materiae** (Topological Materials Database)
 - Standard database of topological materials.
 - Excluded magnetic materials + trivial metals.
- **Features**
 - Are all ignored intentionally except for space group number.
- **Characteristics**
 - 22,245 total materials in our data set
 - 1,839 topological insulators
 - 1,238 topological crystalline insulators
 - 2,713 high-symmetry point semi-metal
 - 2,292 high-symmetry line semi-metal
 - 10,657 Trivial insulators (topology is diagnosed by symmetry)
 - 3,506 Trivial insulators (undiagnosable by symmetry)

SI group $\neq$ Z1



Labelled space

SI group = Z1



Discovery space

nature

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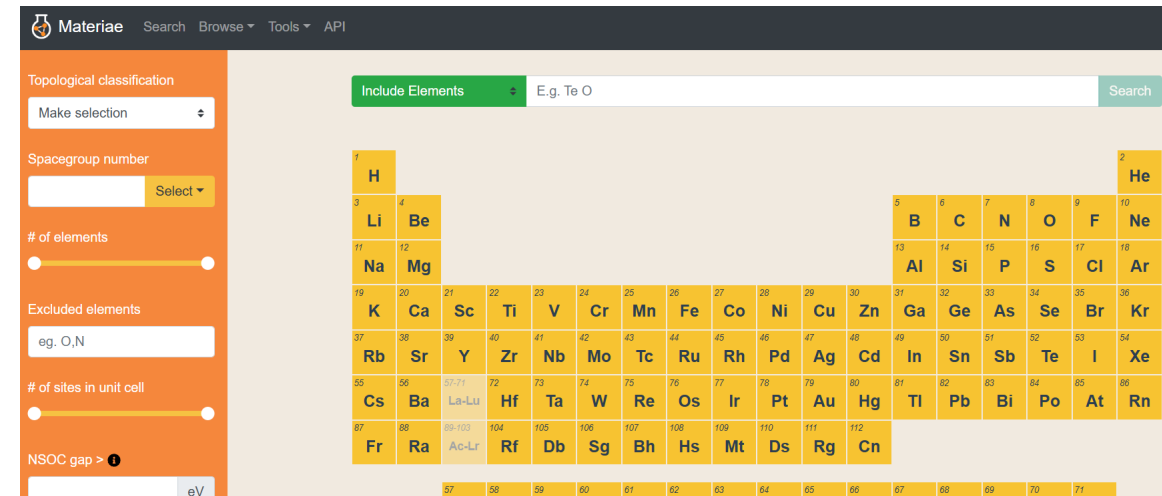
Article | Published: 27 February 2019

Catalogue of topological electronic materials

[Tiantian Zhang](#), [Yi Jiang](#), [Zhida Song](#), [He Huang](#), [Yuqing He](#), [Zhong Fang](#), [Hongming Weng](#)  & [Chen Fang](#) 

[Nature](#) 566, 475–479 (2019) | [Cite this article](#)

40k Accesses | 836 Citations | 46 Altmetric | [Metrics](#)



Po, H.C., Vishwanath, A. & Watanabe, H. Symmetry-based indicators of topological invariants in the 230 space groups. *Nat Commun* 8, 50 (2017). <https://doi.org/10.1038/s41467-017-00133-2>



Training space

Sl group $\neq$ Z1

18,739

Labelled dataset

Topology can be diagnosed with symmetry

10,657

Not Topological*

8,082

Topological

Training & Optimization

Final model

discovery space

Sl group = Z1

3,506

Discovery dataset

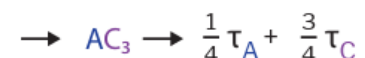
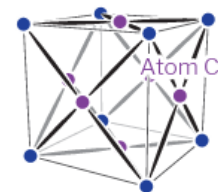
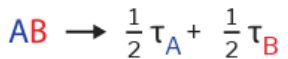
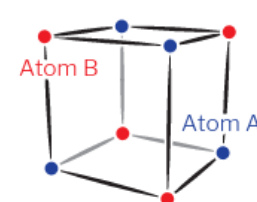
Topology *cannot* be diagnosed with symmetry

Apply final model to find new material candidates

Proposed 'linear' approach in literature

a

Heuristic chemical rule



Heuristic diagnosis

Topologically trivial

<0

Topologically nontrivial

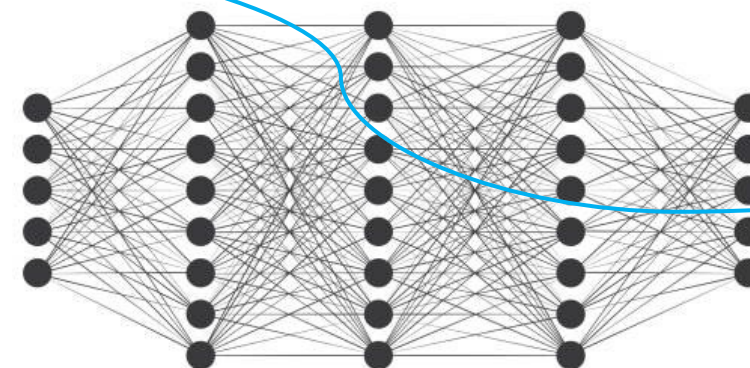
>0

Avg Accuracy: **82.7 %** using nested cross validation

Neural Network approach:

Remove linearity condition:
 $f(A,B,...)=f(A)+f(B)+...$

Chemical Stoichiometry



Probability (topological)

New topological material candidates

Ab initio validation

b High-throughput discovery

All materials to screen

Topogivity prediction

Ab initio validation

Topologically nontrivial materials

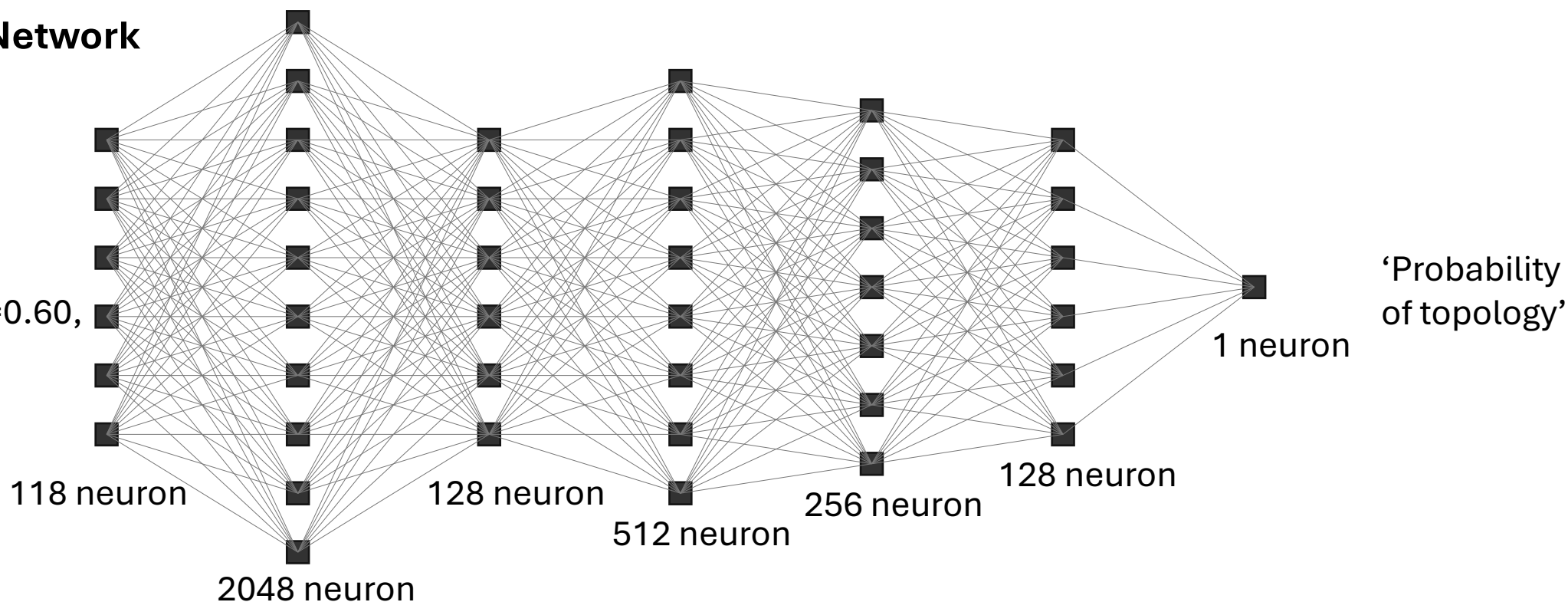
Q1- Can a NN outperforms the literature's method?
Q2- What's the knowledge threshold of uncovering nontrivial topology with only the chemical formula?



ElemNet Neural Network

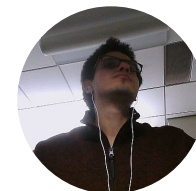
Input example:

$\text{Bi}_2\text{Te}_3 \rightarrow [\text{Bi}=0.40, \text{Te}=0.60, \text{others}=0]$



- **Input:** 118-dimensional vector of *elemental fractions* (no hand-crafted material descriptors).
- **Deep fully-connected network** trained **end-to-end** from composition to property/class.
- **Layer dimensions:**
 $118 \rightarrow 2048 \rightarrow 128 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 1$
- Each hidden layer block consists of:
Linear $\rightarrow$ Batch Normalization $\rightarrow$ LeakyReLU $\rightarrow$ Dropout
- **Batch Normalization** smooths optimization, accelerates training, and stabilizes activation scales across layers.
- **Dropout ($p \approx 0.22$)** reduces overfitting and encourages robust latent chemical representations.
- **Output layer:** Linear $\rightarrow$ **Sigmoid** for binary classification (topological vs trivial).

Model architecture, including activation functions, and dropout were found after



The architecture was chosen using **automated neural architecture search** with **Optuna**, where the model structure was treated as a learnable hyperparameter.

Search Strategy

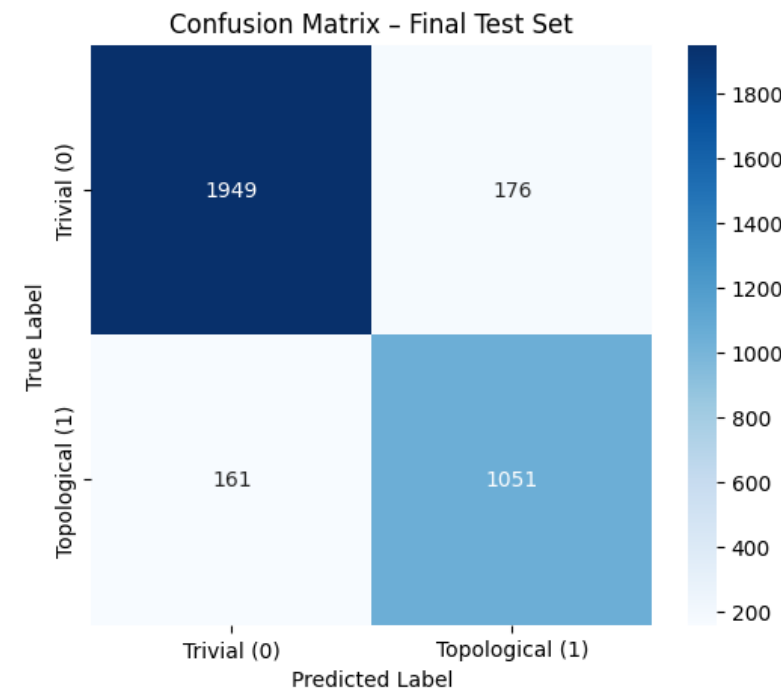
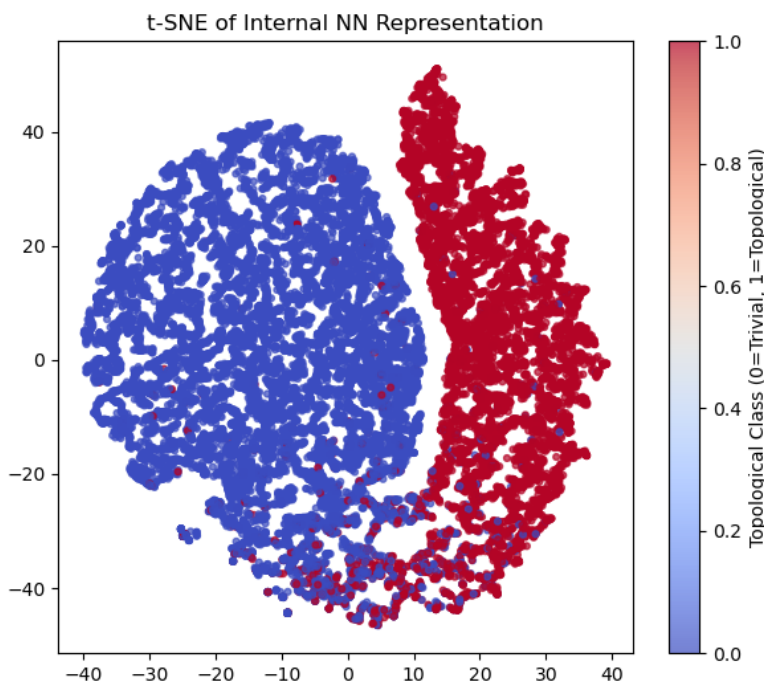
- We defined a **flexible ElemNet model** where:
 - Number of layers (**3–20**) was variable.
 - Each layer size was chosen from {128, 256, 512, 1024, 2048}
 - Dropout probability, learning rate, and activation function were also tuned.
- Optuna explored different architectures using **Bayesian optimization**, guided by validation **ROC-AUC**.
- Each candidate model trained only for **10 warm-up epochs** to estimate performance efficiently.
- A 70-15-15 **train-val-test** split is adopted with training with a 5-fold cross validation.

A100 GPU of Colab is used

Accuracy ~ 90.0% outperforming literature's ML model *.

What did the NN learn ?

The **internal 128-dimensional representation** learned by the network into 2D using t-SNE, the materials ***naturally cluster into two distinct regions*** corresponding to **trivial** and **topological** classes — **without the model being explicitly instructed** to separate them in latent space.



the network has learned a **chemically meaningful feature space**

*Nano Lett. 2023, 23, 3, 772–778



Predicting new topological materials in the discovery space

- Predicted **335** new topological materials with high confidence.
- Among the 69 materials predicted in References 1 & 2, our model predicts 68.

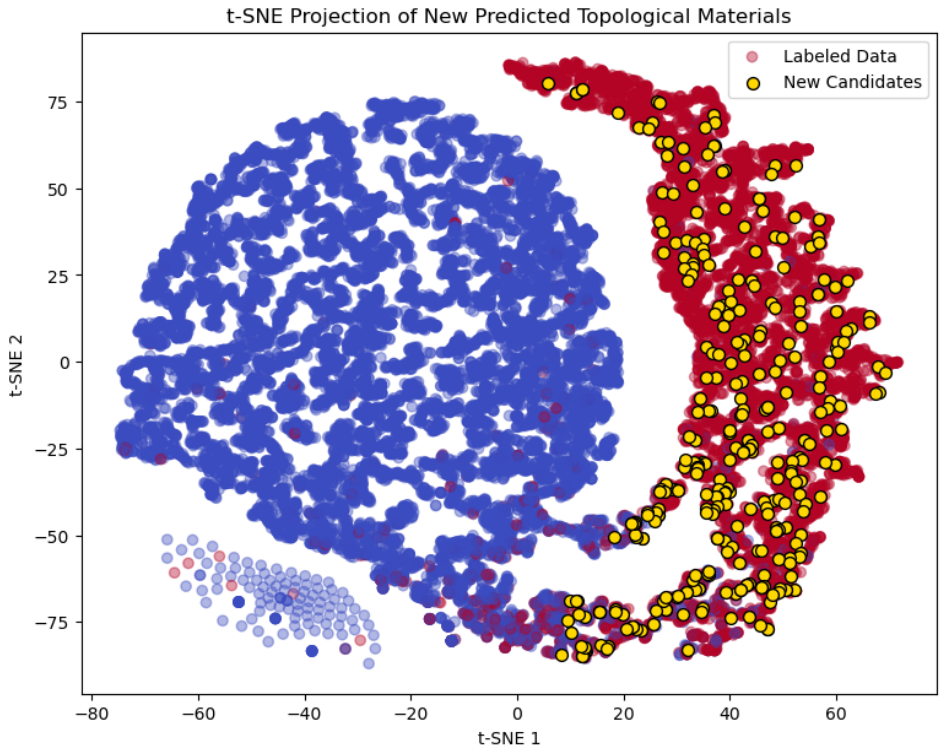
e.g.

Ag3Sb	NaFeS2
BiPd2Pb	Ho10Si17
BiPd2Pb	Nd4Ge7
MgPt	Yb11InSb5
BiPd	TiSi
BiPd	TiSi
CeSiPt	Yb11GaSb
BiPd	Ho2NiGe6
Al2Mo3C	ZrCu
LiFeP	LiIr3
Ca(PIr)2	TmGePd
LiFeAs	Yb2Ge6Pt
CeSiNi	Si2Mo
LiRh3	YbGePd
Sb3Pd8	K2CdSn
Ca2Ni7P4	AsPd2
Sb2Pd5	La2Fe2S5
YCuPb	NdCoSi3
CePIr	TiSiRu
NdCoGe3	CaGe2
SbPd2	Pr2Ni7P4
LaBPt2	LaCoC2
Lu4Zn5Ge	Y3Ni2
Zn3Ga4Pc	Pd3(PbS)2
YbSnAu	DySn3
Tb4Zn5Ge	Si3MoPt2
Th7Ru3	LaGe3Bt
YbSiAu	InAs
TaAs	Ca2GePd2
PrBC	Ni19Ge12
Sc3Re2Si3	Cr2P
	CrWN2
	PrGe3Rh

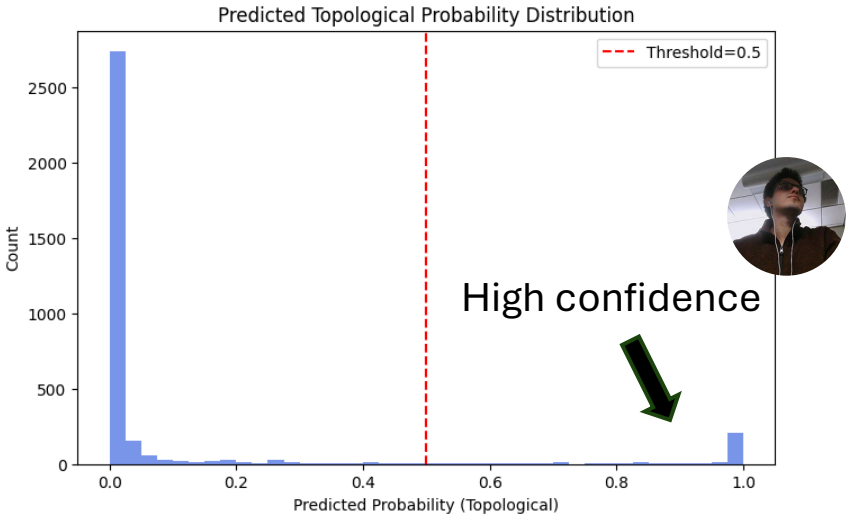
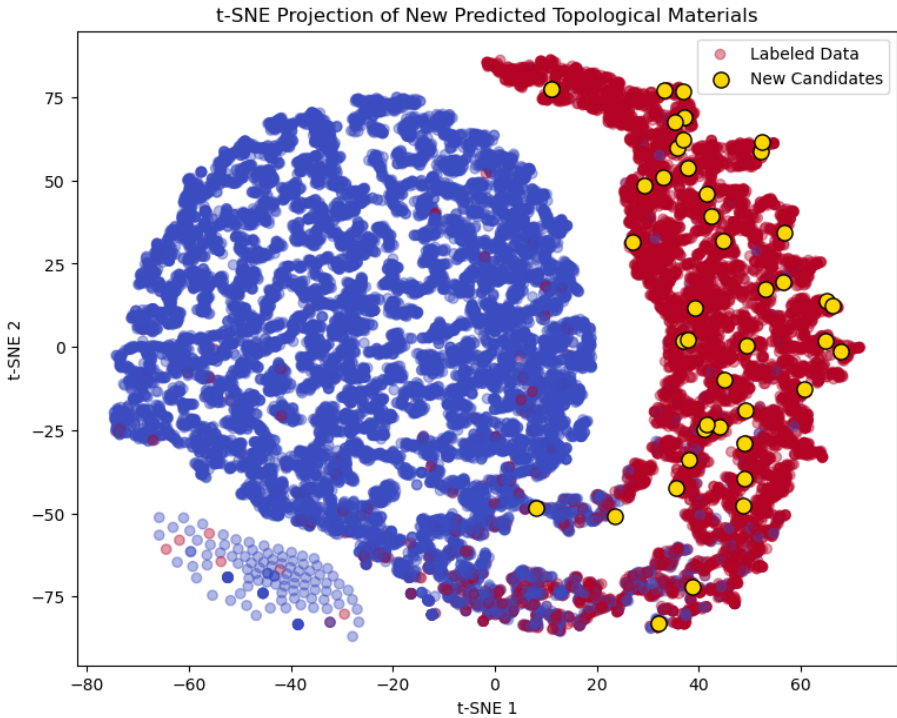


- 61 are confirmed with DFT calculations in Ref 1 & 2.
- Some of these have been confirmed experimentally such as the Weyl semi-metal TaAs using ARPES (Observation of Weyl nodes in TaAs. *Nature Phys* **11**, 724–727)
- 266 materials yet to be confirmed with either DFT or exp.

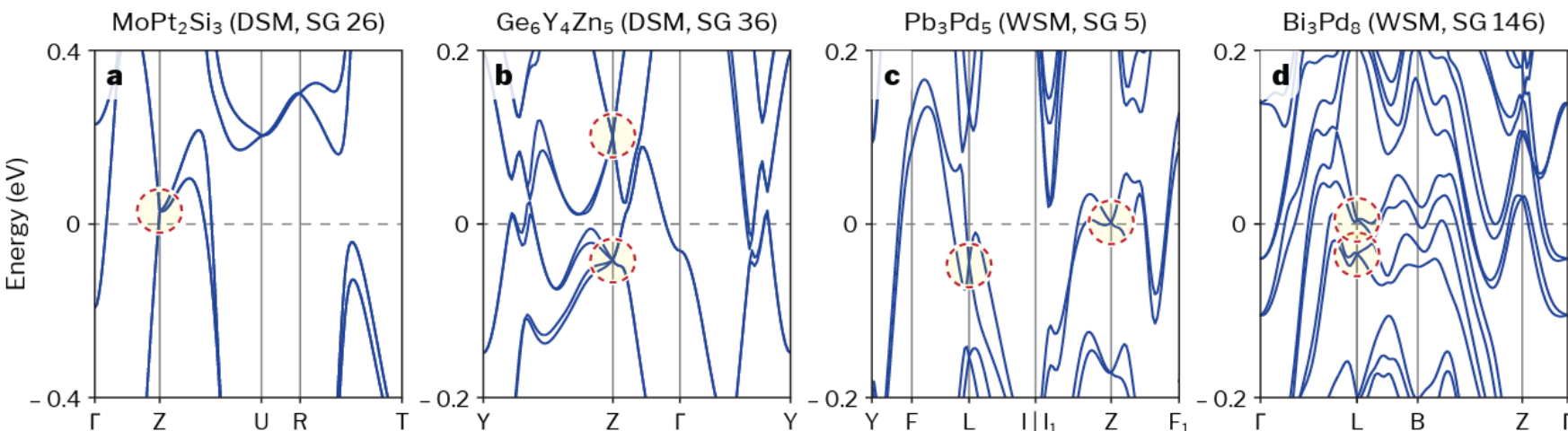
All predicted materials using our NN



Predicted materials in Ref 1

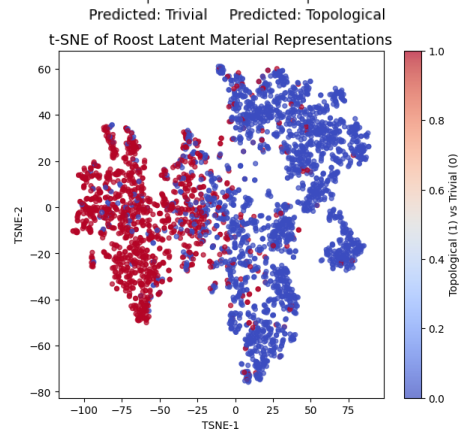
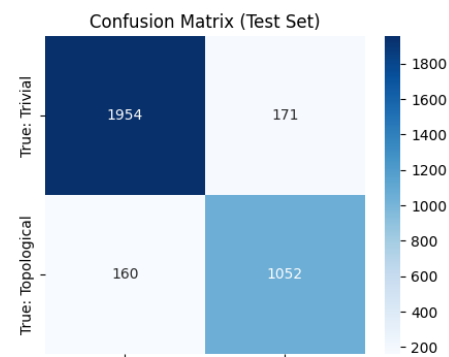
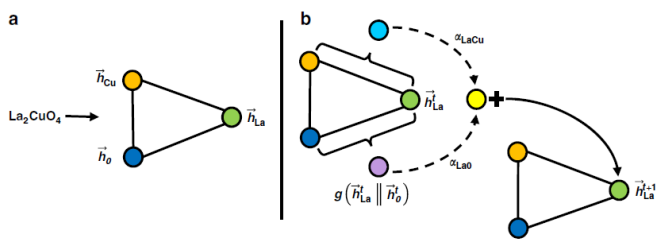


Some DFT calculations confirming topology in a few of the predicted candidate materials:

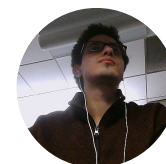
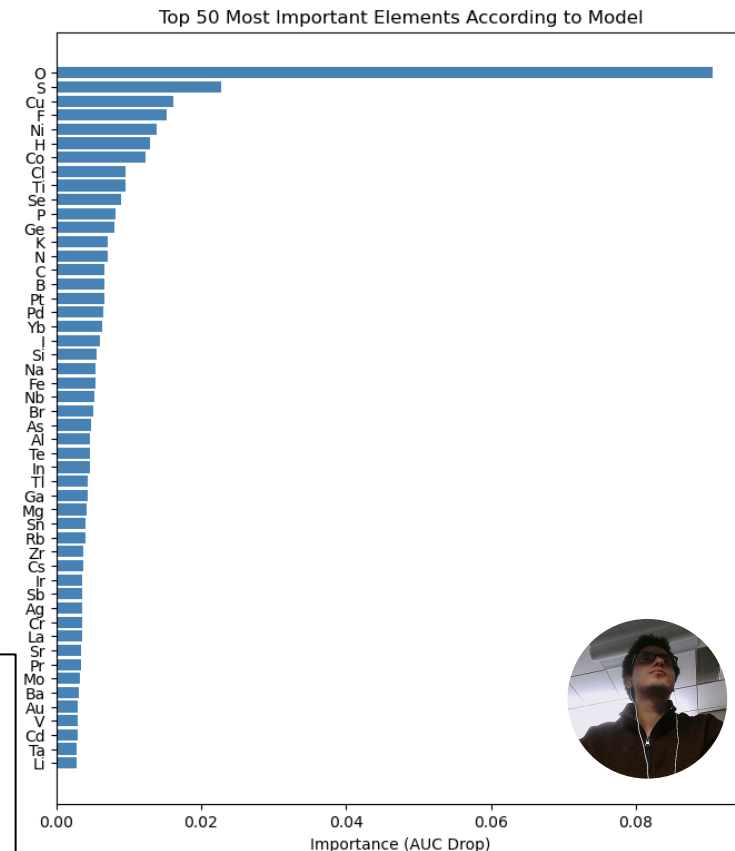


MPNN model:

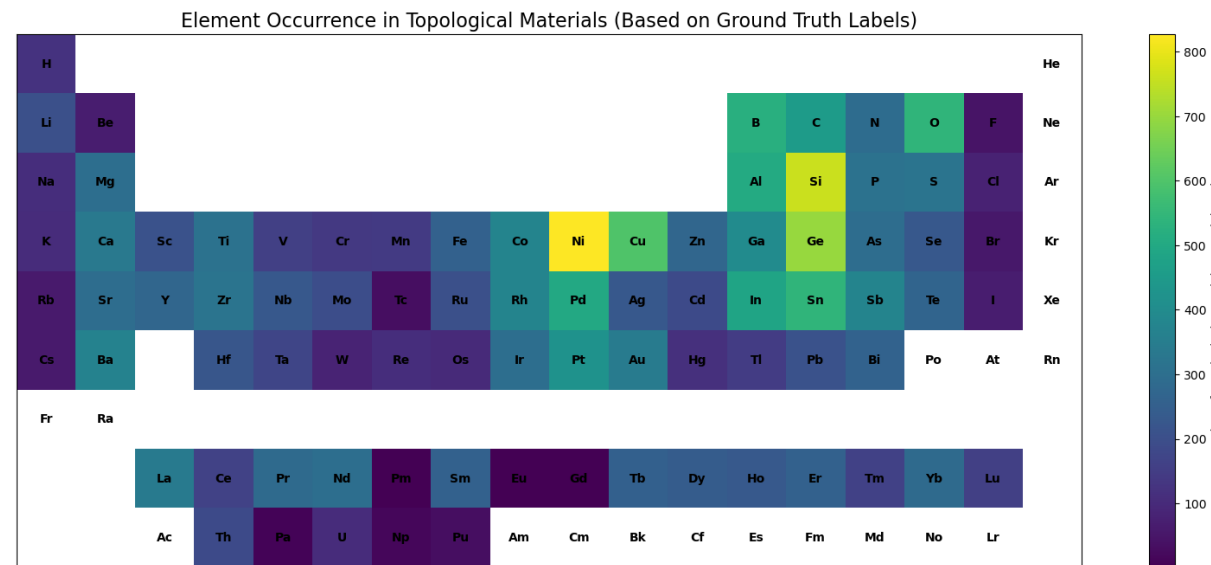
- Each element is mapped to a **learned embedding vector**, capturing periodic-table chemistry (similar to word embeddings in NLP).
- The model builds a **composition graph**, where nodes are elements and edges represent their **stoichiometric relationships**.
- A **message-passing neural network** aggregates interactions between elements to learn **composition-dependent properties**.
- The final global embedding is used for **classification** (topological vs. trivial) via a fully connected prediction head.



Element importance in our NN:



Almost same accuracy ~90%!
(This might be the **threshold** of knowledge given only the chemical stoichiometry)



Summary:

- **Outperformed** recent literature models, mentioned above, **with ~ 90.0% accuracy** compared to the ~ 83.5% in literature, **pushing the conjectured limit of chemical composition-only accurate prediction of topology**, first argued in the first reference mentioned previously.
- **New 266 topological material candidates, not explored before.**
- The topological classification task has indeed successfully learned from composition-only features, demonstrating that chemically driven trends in band topology can be captured without explicit structural or electronic descriptors.
- The attention-based MPNN model achieved a very close result to ElemNet, probably hitting the limit of accuracies reachable with only the chemical stoichiometry.
- Analysis of learned representations suggests that certain elemental groups and combinations exhibit strong correlations with topological phases, consistent with known chemical trends in SOC-driven band inversion and symmetry-protected states.

Thank You!

